



Artificial Intelligence: An introduction

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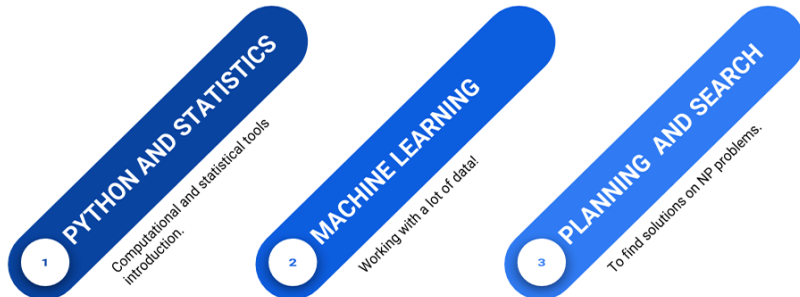


Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

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- ▶ Contenido del curso:
<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

Course outline



- ▶ **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- ▶ **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course



Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

Second Module: Machine Learning

- ▶ Naive Bayes
- ▶ Classification trees
- ▶ Support Vector Machines
- ▶ Non-supervised learning
- ▶ Introduction to Deep-learning



Third module: Planning and Searching

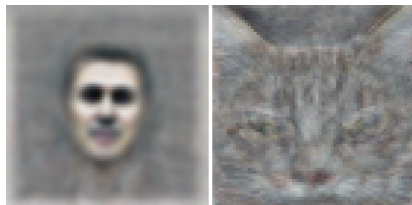
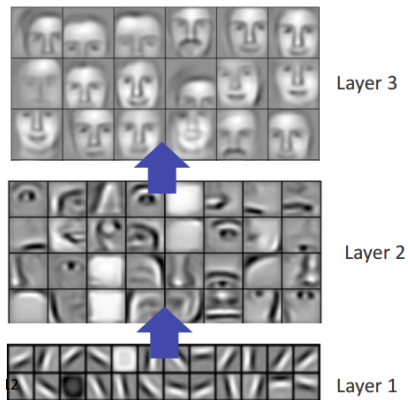
- ▶ Planning and searching
- ▶ Genetic algorithms (GA)
- ▶ Simulated annealing (SA)



Big challenges

Now, algorithms can create new things . . . as abstract as the art! (Netflix 5:15, Bill Nye)

Big challenges



A photograph of a Go match between a human player and the AI program AlphaGo. The human player, wearing a dark suit, is seated on the left, looking at the board. AlphaGo, represented by a white robot arm, is on the right, placing a white stone on the board. The board is a large Go board with black and white stones. The background is dark, and the scene is lit to highlight the players and the board. The text "Video 01" is overlaid in green at the top right. The text "ALPHAGO" is written in large white letters at the bottom.

Video 01

ALPHAGO

Big challenges

Video 02

ALPHAGO

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

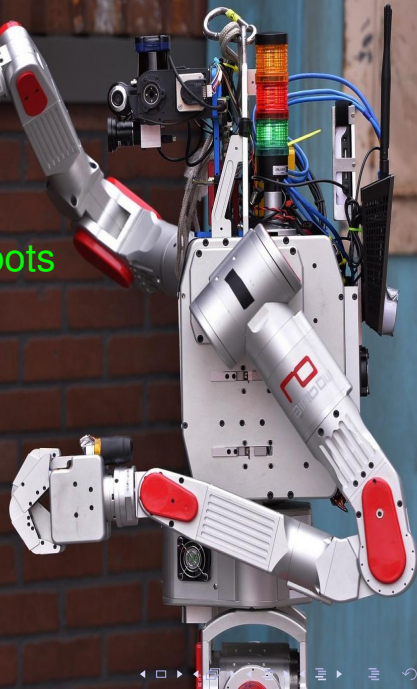


As support for decisions making

- ▶ Medicine
- ▶ Politics
- ▶ Laws



NOVA Rise of the Robots





y ahora ...Atlas. Google again!

Boston Dynamics

DARPA



Structure of the Course

- ▶ 40% Talleres (Problemsets)
- ▶ 30% Parciales (Quizes)
- ▶ 30% Proyecto funcional IA
 - ▶ Solo se evalua el curso, si ha sido previamente aprobado antes del primer corte.

Curso en GITLAB:

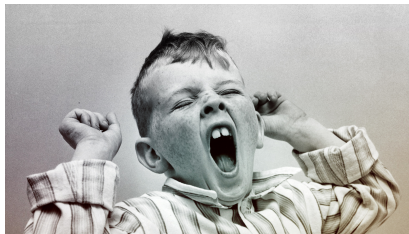
`https://gitlab.com/bivl2ab/academico/
cursos-uis/ai/ai-uis-student`

PREVIOUS PROJECTS

- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md

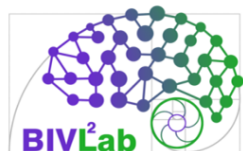


Thank you for your attention ...



... It's time to wake up

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